FINE-GRAINED STYLE MODELLING AND TRANSFER IN TEXT-TO-SPEECH SYNTHESIS VIA CONTENT-STYLE DISENTANGLEMENT

Daxin Tan, Tan Lee

Department of Electronic Engineering, The Chinese University of Hong Kong, Hong Kong

arXiv:2011.03943v1 [eess.AS] 8 Nov 2020

ABSTRACT

This paper presents a novel neural model for fine-grained style modeling and transfer in expressive text-to-speech (TTS) synthesis. By applying collaborative learning and adversarial learning strategies with thoughtfully designed loss functions, the proposed model is able to perform effective phoneme-level disentanglement of content factor and style factor of speech. Speech style transfer can be achieved by combining the style embedding extracted from a reference utterance with the phoneme embedding derived from the source text. Results of objective evaluation show that the synthesized speech preserves the intended content and carries similar prosody to the reference speech. Results of subjective evaluation show that the new model performs better than other fine-grained style transfer TTS models.

***Index Terms*—** speech synthesis, style transfer, prosody

1. INTRODUCTION

Human speech production manifests a complex integration of physical, cognitive and affective processes. The realization of a spoken utterance involves three factors, namely the content factor, the speaker factor and the style factor. The content factor is determined by the linguistic content of speech. The speaker factor refers mainly to voice characteristics that are pertinent to identifying the speaker. While an unambiguous definition of "style" may not exist, in this study the style factor is assumed to cover broadly any aspect of the speech utterance that does not contribute to the acquisition of linguistic content and the identification of speaker [1]. It is commonly agreed that the style of speech is perceptually related to variation of pitch, loudness and duration.

Different tasks of spoken language processing can be viewed as purposeful processes of analyzing, regulating and/or manipulating one or more of the three factors. Automatic speech recognition (ASR) and text-to-speech synthesis (TTS) are focused on the content factor with the speaker and style factors being suppressed or ignored. Speaker recognition and voice conversion aim to capture and manipulate the speaker factor, while emotion recognition and expressive speech synthesis deal primarily with the style factor.

Relating to expressive TTS, speech style transfer (SST) is the process of transferring the style of a reference speech utterance into the content of a source speech utterance. In a typical SST model, a style embedding is obtained from the reference speech, and subsequently combined with the text of source speech (and probably a speaker embedding) to condition speech generation in a neural TTS system. The studies in [2] and [3] extended the Tacotron system with the use of fix-length style embedding at utterance level. Variational auto-encoder (VAE) and hierarchical structure have been applied to improve the representation capability of learned style embeddings in [4], [5] and [6] . To facilitate fine-grained style control on specific parts of an utterance, phoneme-level prosody embedding was investigated in [7] and [8]. In [7], a secondary attention module was used to generate prosody embedding from mel-spectrogram of the reference speech. In [8], style embedding was derived from acoustic features at phoneme level based on HMM forced alignment.

An SST model typically comprises a style encoder and a text encoder. The style encoder aims to encode the style factor of reference speech into a style embedding, while the text encoder generates a content embedding from the text of source speech. The two embeddings can be combined to synthesize speech with the respective content and style. The SST model is trained with speech utterances with corresponding text content. That is, the same utterance serves as both source speech and reference speech. The objective of training is to minimize reconstruction loss of the synthesized speech with respect to the input speech. Since no constraint is exerted to enforce the style encoder to encode exclusively the style information in reference speech, a common problem, referred to as "content leakage" in [9], is that the learned style embedding undesirably captures a significant amount of content information. Such embedding cannot be used for style transfer with varying speech content. In [9], this problem was addressed by minimizing the mutual information between content and style embedding during training. In [10], a pairwise training strategy was proposed to enforce correct mapping from input text to different speech utterances.

In the present study, a novel neural model for effective disentanglement of content and style factors is developed. Collaborative learning and adversarial learning strategies are applied to the encoders to maximize the separation of phoneme-level content and style information. The proposed model can be used to extract variable-length phoneme-level style repre-

sentation from mel-spectrogram of the reference speech and realize fine-grained style modeling and transfer. Results of objective and subjective evaluation show that the proposed model can be used for varying-content style transfer in the single-speaker case, and show noticeable performance advantage against other fine-grained style transfer TTS models.

2. THE PROPOSED MODEL

An overview of the proposed model is shown as in Figure 1. The reference speech is divided into a sequence of phoneme segments, from each of which a content embedding and a style embedding are extracted by the phoneme-segment content-style disentanglement (PS-CSD) module. For speech generation, the extracted style embedding is used jointly with the phoneme embedding derived from the text of source speech. For model training, the same utterance serves as both source speech and reference speech.

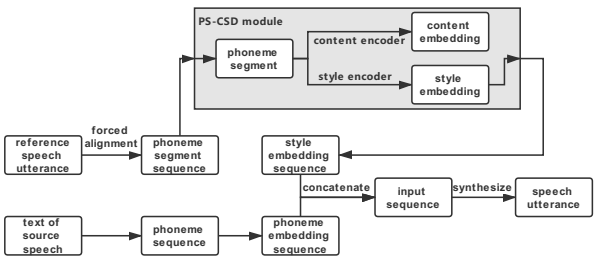

Fig. 1. Overview of the proposed model

2.1. The PS-CSD module

Let u_i denote the ith utterance and t_i be the corresponding text transcription. t_i is expressed in the form of a phoneme sequence, i.e., $t_i = [w_1, ..., w_{m_i}]$, where m_i is the number of phonemes, and w_k denotes the kth phoneme in t_i. By applying forced alignment, with silence/pause segments excluded, u_i is also divided into m_i segments, denoted as $\{s_1, ..., s_{m_i}\}$, and segment s_k corresponds to phoneme w_k. Without considering the temporal relation among segments, the collection of (s_k, w_k) are treated as independent data instances for the training of PS-CSD. s_k could be represented as frame-level acoustic features $[f_1, ..., f_{t_k}]$, where t_k is the number of frames in s_k. Note that w_k is one of the phonemes in concerned language. In this study, we use the 39 English phonemes as defined in the APARBET.

The PS-CSD consists of the following components as shown in Figure 2:

Content encoder (E_c): to encode the content factor of segment s into the content embedding z_c, i.e., $z_c = E_c(s)$

Style encoder (E_s): to encode the style factor of segment s into the style embedding z_s, i.e., $z_s = E_s(s)$

Content-to-phoneme classifier (C_c): to predict the phoneme identity w_c from the content embedding z_c, i.e., $w_c = C_c(z_c)$

Style-to-phoneme classifier (C_s): to predict the phoneme identity w_s from the style embedding z_s, i.e., $w_s = C_s(z_s)$

Decoder (D): to reconstruct segment s' from the content embedding z_c and the style embedding z_s, i.e., $s' = D(z_c, z_s)$

Segment classifier (C_{seg}): to determine if a speech segment is from natural speech (true) or synthesized (false), i.e., $P_{true} = C_{seg}(\{s, s'\}))$

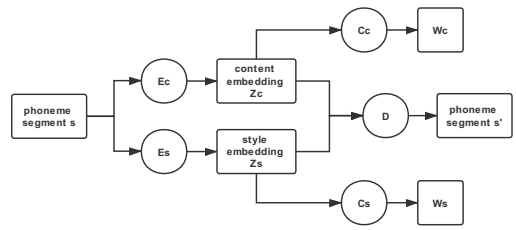

Fig. 2. Detailed design of the PS-CSD module

The training strategies for PS-CSD are summarized as in Table 1 and explained in the following subsections.

Table 1. PS-CSD module training algorithm

Input: segments and corresponding phonemes: (s_k, w_k)
Repeat until convergence:
1. Train E_c, E_s, D by minimizing Eq.(1)
2. Train E_c, C_c by minimizing Eq.(2) and Eq.(3)
3. Fix E_s, train C_s by minimizing Eq.(4)
4. Fix C_s, train E_s by minimizing Eq.(5)
5. Fix E_c, E_s, D, train C_{seg} by minimizing Eq.(6)
6. Fix C_{seg}, train E_c, E_s, D by minimizing Eq.(7)

2.1.1. *Auto-encoder training*

The content encoder E_c, the style encoder E_s and the decoder D together make up an auto-encoder framework. The difference between the reconstructed segment and the original one is measured by the $L2$ norm of their short-time spectral features, referred to as the spectrogram loss L_s, and the binary cross-entropy loss of the gates L_g, which indicates the segment lengths [11]. For decoder training, ground-truth acoustic features from the current frame are concatenated with the content embedding and the style embedding to predict the next frame. The overall auto-encoder loss L_{auto} is given as,

$$L_{auto}(\theta_{E_c}, \theta_{E_s}, \theta_D) = \sum_{s_k} \sum_{L_s, L_g} L(s_k, D(E_c(s_k), E_s(s_k))) \tag{1}$$

where θ denotes the model parameters to be optimized.

Basic auto-encoder training is not expected to achieve the desired purpose of disentangling content and style factors, because the roles of the two encoders are not differentiated. As described below, collaborative training is suggested for training the content encoder while adversarial training is applied to the style encoder.

2.1.2. *Collaborative training of content encoder*

The content embedding is expected to carry pertinent information to phoneme identification. Thus an auxiliary content-

to-phoneme classifier is introduced to assess the goodness of content embedding. By training the content encoder and the content-to-phoneme classifier collaboratively, the content embedding is forced to capture phoneme identity information. The content-to-phoneme classification loss is defined as,

$$L_c(\theta_{E_c}, \theta_{C_c}) = \sum_{(s_k, w_k)} -\log P(w_k | C_c(E_c(s_k))). \quad (2)$$

To ensure that content embeddings extracted from different segments of the same phoneme are similar, the following contrast loss L_{contra} is imposed,

$$L_{contra}(\theta_{E_c}) = \sum_{\substack{(s_i, w_i), \\ (s_j, w_j)}} \mathbb{1}_{w_i = w_j} ||E_c(s_i) - E_c(s_j)||_2 \quad (3)$$

2.1.3. *Adversarial training of style encoder*

In contrast to the content embedding, the style embedding is desired not to encode any information about phoneme identity, or that the phoneme carried by a segment should be non-identifiable from its style embedding. This objective is achieved via adversarial training [12]. In the discrimination phase, the parameters of style encoder are fixed, and the style-to-phoneme classifier is trained to perform phoneme identification from the style embedding,

$$L_s^{dis}(\theta_{C_s}) = \sum_{(s_k, w_k)} -\log P(w_k | C_s(E_s(s_k))) \quad (4)$$

In the generation phase, the parameters of style-to-phoneme classifier are fixed, and the style encoder is trained such that the segment's phoneme identity cannot be predicted from the style embedding. The following loss is defined to enforce equal posterior probabilities across all phonemes,

$$L_s^{gen}(\theta_{E_s}) = \sum_{s_k} \sum_{w \in N_w} ||P(w | C_s(E_s(s_k))) - \frac{1}{N_w}||_2 \quad (5)$$

where N_w denotes the total number of phonemes.

2.1.4. *Adversarial enhancement*

An adversarial enhancement process is applied to supplement the model's reconstruction function. A discriminator network C_{seg} is utilized to judge if a given segment is from natural speech or synthesized, while the auto-encoder serves as a generator model to confuse the discriminator. For the discriminator, we have, $s'_k = D(E_c(s_k), E_s(s_k))$,

$$L_{seg}^{dis}(\theta_{C_{seg}}) = \sum_{s_k} -[\log C_{seg}(s_k) + \log(1 - C_{seg}(s'_k)] \quad (6)$$

For the generator, the loss is given as,

$$L_{seg}^{gen}(\theta_{E_c}, \theta_{E_s}, \theta_D) = \sum_{s_k} -[\log C_{seg}(s'_k) + \log(1 - C_{seg}(s_k)] \quad (7)$$

2.2. Utterance-level synthesis

Upon completion of the training of PS-CSD, the system in Figure 1 is trained to perform speech synthesis at utterance level. It is similar to the standard Tacotron 2 model [11], except that the input comprises a phoneme sequence translated from the input text and a style embedding sequence generated by PS-CSD. The phoneme sequence is first converted to a phoneme embedding sequence using a trainable look-up table, which is subsequently combined with the style embedding sequence in a phoneme-by-phoneme manner. For model training, both the phoneme sequence and the style embedding sequence are obtained from the same utterance. For SST, the phoneme sequence corresponds to a source utterance and the style embedding sequence is from a reference utterance, which generally has different content from the source utterance. In this way, the synthesized speech would carry the source speech content and the reference speech style.

3. EXPERIMENTS

The LJ Speech dataset [13] is used in this study. It contains $13,100$ utterances from a single speaker. Their total length is about 24 hours. 90% of the utterances are used as training data and the remaining 10% as test data. Forced alignment of speech is carried out using the Montreal Forced Aligner [14] based on the CMU Pronouncing Dictionary.

Frame-level acoustic features are obtained in a similar setting to the standard Tacotron 2 model. In PS-CSD, both the content encoder and the style encoder are implemented as a bidirectional LSTM of dimension 512 (256 in each direction), in which the last cell state is projected to embedding via a linear layer. The decoder contains a unidirectional LSTM of dimension 512, with two linear layers applied on the output at each time step. One of them predicts the acoustic features and the other indicates the end of sequence. The dimensions of content embedding and style embedding are both 64. For utterance-level speech synthesis, the dimension of phoneme embedding is 64. The WaveGlow vocoder [15] is used to generate speech waveform from the predicted mel-spectrogram. Audio samples are available at our demo page[1].

4. RESULTS AND DISCUSSION

Objective and subjective evaluation are carried out on the proposed SST model in comparison with two recent SST systems described in [7] and [8].

4.1. Objective evaluation

4.1.1. *Reconstruction of speech*

Being able to reconstructing an input utterance from the disentangled content and style embeddings is a basic require-

[1] https://daxintan-cuhk.github.io/ps-csd-speech

ment for the encoder-decoder model. In this case, the source speech and the reference speech are from the same test utterance. The similarity between the synthesized speech and the original input speech is used to indicate the performance of reconstruction. Following [2], Dynamic Time Warping (DTW) based on MFCC features is first performed to align the two utterances, and a set of frame-level error or distortion metrics are computed. As shown in Table 2, these metrics include the Voicing Decision Error (VDE), Gross Pitch Error (GPE), F0 Frame Error (FFE) and Mel Cepstral Distortion (MCD). The proposed model shows slightly better or comparable performance than the systems reported in [7] and [8].

Table 2. Objective evaluation on speech reconstruction

	VDE (%)	GPE (%)	FFE (%)	MCD_{13}
Lee & Kim [7]	9.05	8.16	13.04	10.91
Klimkov *et al.* [8]	12.00	5.43	14.53	11.67
Our model	11.03	4.57	13.15	10.49

4.1.2. Recombination of content and style embeddings

When the proposed model is used for style transfer, the reference speech for deriving style embedding sequence is different from the source speech which specifies the text content (phoneme sequence). This is referred to as recombination. In the present study, the source speech and the reference speech are required to contain the same number of phonemes. The synthesized utterance is expected to contain the same content as the source speech. The content similarity is assessed using a pre-trained ASR model. In our experiment, an end-to-end ASR system provided in the ESPnet toolkit [16] is used. The word error rate (WER) and phone error rate (PER) of synthesized speech are evaluated with respect to the source speech transcription. The system in [7] is generally unable to retain the source content, i.e., the synthesized speech follows the reference speech in both content and style. Compared with the system of [8], our proposed model is significantly better in preserving the source content. The comparison is shown as in Table 3.

Table 3. ASR error rates on style-changed speech

Metric	WER (%)	PER (%)
Lee & Kim [7]	90.4	74.1
Klimkov *et al.* [8]	29.2	14.5
Our model	21.4	8.5

Figure 3 provides an example case of content-style recombination that illustrates the effect of style transfer. The top pane shows the spectrograms and pitch contours of two different utterances of natural speech. The middle pane shows the reconstructed utterances, i.e., synthesized with both phoneme embedding (content) and style embedings coming from the same utterance. The bottom pane shows the results of speech generation with the two utterances' phoneme embeddings (content) swapped. The utterances synthesized with the same style embedding, i.e., those in the same column of the figure, show highly similar patterns of temporal pitch variation. The most notable similarities are marked by the colored boxes.

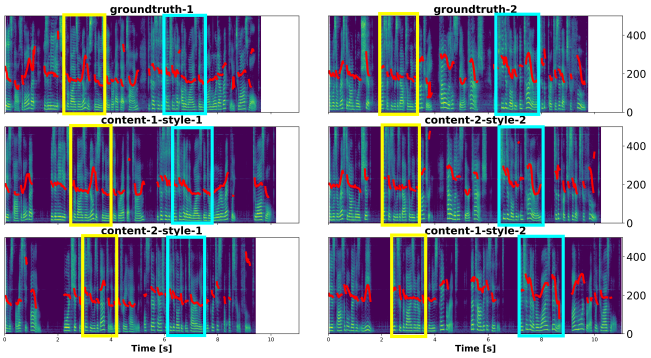

Fig. 3. Spectrograms and pitch contours of two test utterances of natural speech and synthesized speech with different combinations of content and style embeddings.

4.2. Subjective evaluation

Subjective listening tests were carried out to evaluate the reconstructed speech and the style-transferred speech separately. For evaluating the performance of speech reconstruction, the listeners are required to rate overall similarity between synthesized speech and source speech (natural speech), For evaluating style transfer, separate ratings are given on the content similarity between the synthesized speech and the source speech and the style similarity between the synthesized speech and the reference speech. The score of rating ranges from 0 (completely different) to 5 (exactly the same). A total of 30 listeners participate in the listening test, which were carried out via the Amazon Mechanical Turk platform. Each listener was required to evaluate 30 sets of test utterances in both cases.

Table 4. Subjective evaluation results

MOS	Reconstruction	Recombination	
	Overall	Content	Style
Lee & Kim [7]	2.09	-	-
Klimkov *et al.* [8]	2.95	2.97	2.53
Our model	3.47	3.41	2.54

Table 4 shows the results of subjective evaluation. The system of [7] is not able to perform varying-content style transfer. Our proposed model attains a score of 3.47 on reconstructed speech, versus 2.95 and 2.09 by the existing systems. For speech generation with style transfer, our model obtained similar rating to the system in [8] in the aspect of style and significantly higher rating in content preservation.

5. ACKNOWLEDGEMENT

This research is partially supported by a Tier 3 funding from ITSP (Ref: ITS/309/18) of the Hong Kong SAR Government, and a Knowledge Transfer Project Fund (Ref: KPF20QEP26) from the Chinese University of Hong Kong.

6. REFERENCES

[1] Rui Liu, Berrak Sisman, Guanglai Gao, and Haizhou Li, "Expressive tts training with frame and style reconstruction loss," *arXiv preprint arXiv:2008.01490*, 2020.

[2] RJ Skerry-Ryan, Eric Battenberg, Ying Xiao, Yuxuan Wang, Daisy Stanton, Joel Shor, Ron J Weiss, Rob Clark, and Rif A Saurous, "Towards end-to-end prosody transfer for expressive speech synthesis with tacotron," *arXiv preprint arXiv:1803.09047*, 2018.

[3] Yuxuan Wang, Daisy Stanton, Yu Zhang, RJ Skerry-Ryan, Eric Battenberg, Joel Shor, Ying Xiao, Fei Ren, Ye Jia, and Rif A Saurous, "Style tokens: Unsupervised style modeling, control and transfer in end-to-end speech synthesis," *arXiv preprint arXiv:1803.09017*, 2018.

[4] Ya-Jie Zhang, Shifeng Pan, Lei He, and Zhen-Hua Ling, "Learning latent representations for style control and transfer in end-to-end speech synthesis," in *ICASSP 2019-2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2019, pp. 6945–6949.

[5] Tom Kenter, Vincent Wan, Chun-An Chan, Rob Clark, and Jakub Vit, "Chive: Varying prosody in speech synthesis with a linguistically driven dynamic hierarchical conditional variational network," in *International Conference on Machine Learning*, 2019, pp. 3331–3340.

[6] Wei-Ning Hsu, Yu Zhang, Ron J Weiss, Heiga Zen, Yonghui Wu, Yuxuan Wang, Yuan Cao, Ye Jia, Zhifeng Chen, Jonathan Shen, et al., "Hierarchical generative modeling for controllable speech synthesis," *arXiv preprint arXiv:1810.07217*, 2018.

[7] Younggun Lee and Taesu Kim, "Robust and fine-grained prosody control of end-to-end speech synthesis," in *ICASSP 2019-2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2019, pp. 5911–5915.

[8] Viacheslav Klimkov, Srikanth Ronanki, Jonas Rohnke, and Thomas Drugman, "Fine-grained robust prosody transfer for single-speaker neural text-to-speech," *arXiv preprint arXiv:1907.02479*, 2019.

[9] Ting-Yao Hu, Ashish Shrivastava, Oncel Tuzel, and Chandra Dhir, "Unsupervised style and content separation by minimizing mutual information for speech synthesis," in *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2020, pp. 3267–3271.

[10] Shuang Ma, Daniel Mcduff, and Yale Song, "Neural tts stylization with adversarial and collaborative games," in *International Conference on Learning Representations*, 2018.

[11] Jonathan Shen, Ruoming Pang, Ron J Weiss, Mike Schuster, Navdeep Jaitly, Zongheng Yang, Zhifeng Chen, Yu Zhang, Yuxuan Wang, Rj Skerrv-Ryan, et al., "Natural tts synthesis by conditioning wavenet on mel spectrogram predictions," in *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2018, pp. 4779–4783.

[12] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio, "Generative adversarial nets," in *Advances in neural information processing systems*, 2014, pp. 2672–2680.

[13] Keith Ito and Linda Johnson, "The lj speech dataset," https://keithito.com/LJ-Speech-Dataset/, 2017.

[14] Michael McAuliffe, Michaela Socolof, Sarah Mihuc, Michael Wagner, and Morgan Sonderegger, "Montreal forced aligner: Trainable text-speech alignment using kaldi.," in *Interspeech*, 2017, pp. 498–502.

[15] Ryan Prenger, Rafael Valle, and Bryan Catanzaro, "Waveglow: A flow-based generative network for speech synthesis," in *ICASSP 2019-2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2019, pp. 3617–3621.

[16] Shinji Watanabe, Takaaki Hori, Shigeki Karita, Tomoki Hayashi, Jiro Nishitoba, Yuya Unno, Nelson Enrique Yalta Soplin, Jahn Heymann, Matthew Wiesner, Nanxin Chen, et al., "Espnet: End-to-end speech processing toolkit," *arXiv preprint arXiv:1804.00015*, 2018.